

Temporal Superposition in GPT-2: How Temporal Time is Encoded and Distributed Across Layers

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Abstract

We investigate how GPT-2 Small encodes temporal concepts in its residual stream, combining principal component analysis of raw activations with a layer-wise sparse autoencoder (SAE) ablation scan across all 12 layers. Using a 100-prompt historical year corpus spanning roughly 2,800 years and three cyclic concept datasets (days of the week, months of the year, and days of the month), we characterise both linear and circular temporal representations and track how they evolve with depth.

For linear historical time, the temporal axis is strongly present ($|r|_{\text{base}} \approx 0.85\text{--}0.92$) from early layers onward, but the representation transitions from concentrated to distributed: at layers 0–1 a single SAE feature accounts for a drop of ≈ 0.43 in temporal correlation when ablated, whereas by layer 7 no individual feature is load-bearing. We show this early concentration is partly an artifact of BC/AD surface tokens, whose removal collapses the layer-1 signal entirely and shifts peak concentration to layers 4–5, after which both corpora converge to the same fully distributed superposition regime.

For cyclic concepts, we replicate the circular PC2/PC3 geometry reported by prior work for days of the week ($r_{\text{angular}} = 0.662$) and months of the year ($r_{\text{angular}} = 0.748$), and extend the analysis to days of the month, finding a weaker but present circular structure ($r_{\text{angular}} = 0.495$) whose strength is highly prompt-frame dependent. Together, these results suggest that GPT-2 Small encodes linear and cyclic temporal structure through fundamentally different geometric mechanisms, and that surface token identity can dominate early-layer probing signals in ways that obscure genuine semantic encoding.

1 Introduction

Large language models have demonstrated their ability to "know" historical facts, but we have limited understanding of how that knowledge is stored internally. Understanding the geometric structure of factual knowledge in LLMs is a foundational question for mechanistic interpretability.

2 Related Work

TODO

3 Background

3.1 Residual Stream

Transformer models process tokens by passing representations through a sequence of attention and MLP layers. These layers write incremental updates into a shared vector called the residual stream. Formally if $\mathbf{x}^{(\ell)} \in \mathbb{R}^d$ denotes the residual stream at layer ℓ , each layer computes $\mathbf{x}^{(\ell+1)} = \mathbf{x}^{(\ell)} + f^{(\ell)}$, where $f^{(\ell)}$ is the combined attention and MLP transformation. This additive structure means the residual stream accumulates information across depth, and that representations at any layer can be read as a type of superposition of all contributions made so far. In this work we probe the residual stream of GPT-2 Small Radford et al. [2019], a 117 million parameter transformer with a hidden dimension of $d = 768$.

3.2 Superposition

A naive expectation is that each dimension of the residual stream encodes one interpretable feature. In practice, the number of concepts a model must represent exceeds its hidden dimension d ($\mathbb{R}^{768}$ in this paper). As a result, models exploit the fact that high dimensional spaces can accommodate many nearly-orthogonal directions. This phenomenon is called superposition and it allows a model to store a large amount of features at the same time. Features in superposition are distributed across many neurons.

3.3 Sparse Autoencoders

Sparse autoencoders (SAEs) offer a way to recover the underlying precise features from an ambiguous residual stream Cunningham et al. [2024]. An SAE consists of an encoder that projects the d -dimensional activation into a much larger latent space $\mathbb{R}^{d_{\text{SAE}}}$ with $d_{\text{SAE}} \gg d$, followed by a ReLU that enforces sparsity and a linear decoder that reconstructs the original activation. Training minimises reconstruction error subject to an L_1 penalty on the latent activations, encouraging each input to be explained by only a small number of active features at a time. The resulting latent dimensions have been shown to correspond to interpretable, monosemantic concepts Cunningham et al. [2024], Bloom [2024]. In this work we use the SAE of Bloom [2024] trained on GPT-2 Small’s residual stream.

4 Method

4.1 Model and SAE

We do all experiments on GPT-2 Small Radford et al. [2019], a decoder-only transformer with 12 layers, 12 attention heads, and a residual stream dimension of $d = 768$, totalling approximately 117 million parameters. GPT-2 Small is a well-studied model in the context of mechanistic interpretability, making it a good model for probing internal representations.

To decompose residual stream activations, we use the suite of SAEs released by Bloom [2024], trained on GPT-2 Small’s residual stream under the release identifier `gpt2-small-res-jb`. Each SAE is trained on the `hook_resid_pre` activation at its corresponding layer – that is, the residual stream state *before* the attention and MLP transformations of that layer

have been applied. The SAE architecture uses a linear encoder with ReLU activation followed by a linear decoder, with an expansion factor of $64\times$, giving a latent dimension of $d_{\text{SAE}} = 49,152$. For each layer ℓ we load the corresponding SAE independently, allowing us to probe how feature representations evolve across depth.

All activations are extracted at the final token position of each prompt, as this is the position at which the model has processed the full input and is most likely to have integrated contextual information into a coherent representation.

4.2 Prompt Design

All prompts are short declarative sentences placing the concept of interest in natural linguistic context. The model processes each prompt without instruction or chat formatting, with activations extracted at the final token position.

Historical year corpus. The primary dataset consists of 100 prompts spanning approximately 2800 years of history, with 20 prompts per epoch category: **ancient_bc** (−776 to −31), **ancient_ad** (79–529), **medieval**, **early_modern**, and **modern** (through 1991). Each prompt is a single declarative sentence describing a well-known historical event at a specific year, for example “*The first Olympic Games were held in 776 BC*” or “*The Norman Conquest of England occurred in 1066*”. The balanced design ensures equal representation across epochs for ANOVA-based feature selection. BC and AD year ranges are non-overlapping by design — BC items span −776 to −31 and AD items span 79 to 529 — avoiding any ambiguity in numeric year assignment.

Surface token comparison. The superposition depth scan is run twice on the same 50 events to isolate the effect of BC/AD surface tokens. In the first scan, the 40 ancient prompts carry explicit era markers (e.g. “*...in 776 BC*”, “*...in 79 AD*”). In the second scan, era markers are stripped from all ancient prompts, leaving bare year numbers (e.g. “*...in 776*”). Medieval through modern prompts are word-for-word identical across both scans. This design ensures the only variable between the two conditions is the presence or absence of the tokens “BC” and “AD”, with corpus size, event selection, year values, and category labels held constant.

Cyclic concept datasets. To examine whether cyclic temporal concepts are encoded with circular geometry, we construct three datasets. Days of the week ($N = 21$) and months of the year ($N = 36$) each represent their respective concepts across three prompt frames per item: a simple statement (“*Today is Monday*”), a wraparound sequential statement (“*The day after Sunday is Monday*”), and a positional statement (“*Monday is the first day of the working week*”). Days of the month ($N = 93$) extends this structure to all 31 days. The multi-frame design ensures geometric structure in the representations reflects the underlying concept rather than a single surface phrasing.

4.3 Temporal Axis Measurement: Linear Concepts

To quantify how strongly a layer encodes temporal order for historical year prompts, we measure the alignment between the model’s internal representations and chronological position. For each layer l , we pass the year prompt activations through the pretrained SAE

encoder and reconstruct them via the decoder:

$$\hat{x}_i = \mathbf{W}_{\text{dec}} \mathbf{a}_i \in \mathbb{R}^{768}$$

where $\mathbf{a}_i \in \mathbb{R}^{49152}$ is the sparse feature activation vector for prompt i . This reconstruction discards variance not captured by the SAE, meaning our metric reflects temporal structure as expressed through the SAE’s feature basis specifically.

We apply Principal Component Analysis (PCA) to the $(N \times 768)$ reconstruction matrix, retaining a single component (PC1), and compute the absolute Pearson correlation between PC1 scores and numeric year values:

$$|r| = \left| \text{corr} \left(\text{PC1}(\hat{\mathbf{X}}), \mathbf{y} \right) \right|$$

where $y \in \mathbb{R}^N$ contains the numeric year for each prompt, with pre-common-era dates represented as negative integers. We use PC1 because it captures the direction of maximum variance in the reconstruction. If the model encodes temporal position as a linear axis, that axis should dominate the variance structure and align with PC1. The absolute value discards PCA’s arbitrary sign convention. We refer to this quantity as the baseline temporal correlation $|r|_{\text{base}}$ at a given layer, and use drops in $|r|$ under feature ablation as the primary signal in our superposition analysis. described in Section 4.5.

4.4 Temporal Axis Measurement: Cyclic Concepts ****NOTE** May not use this**

For cyclic concepts – days of the week, months of the year, and days of the month – a linear correlation metric is inappropriate. Day 7 and day 1 are adjacent in the cyclic ordering but greatly separated under any linear projection, meaning Pearson r will be near zero even for a geometrically well-structured representation. We therefore adopt a circular measurement pipeline.

We retain five PCA components from the SAE reconstruction and fit an algebraic circle to the PC2/PC3 plane using the Coope(1993) least squares method Coope [1993]. PC1 is deliberately excluded as it captures non-cyclic variance. PC1 captures whether the text refers to a weekday or month and the PC2/PC3 captures the circular structure. For each prompt we compute its angular position relative to the fitted circle center (c_x, c_y) :

$$\theta_i = \arctan 2 (\text{PC3}_i - c_y, \text{PC2}_i - c_x)$$

We then compute the Fisher circular-linear correlation between angular position and the integer label $\mathbf{y}$. A standard Pearson correlation is inappropriate here because θ_i is a circular variable — day 7 and day 1 are adjacent in the cyclic ordering but nearly π apart in angle, causing Pearson r to underestimate the true correspondence. The Fisher metric instead projects the angular variable onto $\cos \theta$ and $\sin \theta$ and measures how well both components jointly predict the label, correctly treating the concept as periodic.

$$r_{\text{angular}} = \sqrt{\frac{r_{xc}^2 + r_{xs}^2 - 2r_{xc}r_{xs}r_{cs}}{1 - r_{cs}^2}}$$

where $r_{xc} = \text{corr}(\cos \theta, \mathbf{y})$, $r_{xs} = \text{corr}(\sin \theta, \mathbf{y})$, and $r_{cs} = \text{corr}(\cos \theta, \sin \theta)$. This metric is bounded in $[0, 1]$ and correctly treats the label as periodic, with $r_{\text{angular}} = 1$ indicating that

the label perfectly predicts angular position on the circle. For the ablation scan in Phase 5, drops in r_{angular} under single-feature ablation replace drops in $|r|$ as the signal metric, with all other aspects of the pipeline held constant.

4.5 Feature Clustering

Given the high dimensionality of the SAE feature space ($d_{\text{SAE}} = 49,152$), directly ablating all features is computationally unmanageable and statistically uninformative. We therefore apply a two stage filtering procedure to identify a coherent subset of features likely to be encoding temporal structure, which then forms the candidate set for the ablation scan.

We first filter to *active* features, those that fire on at least two prompts in the input set. For each active feature f_i we compute a one-way ANOVA F-score across epoch category groups, where each group contains the feature activations for all prompts belonging to that epoch. We retain the top $N_{\text{disc}} = 100$ features by F-score as the discriminative candidate set. However, since two features can both score highly on ANOVA while activating in entirely different patterns across epochs, we apply a second stage to enforce coherence. For each candidate feature we compute its epoch-mean activation profile $\mathbf{p}_i \in \mathbb{R}^K$, z scored across the K epoch categories. Features whose profiles correlate above $|\rho| > 0.5$ are connected in a graph, and connected components of size ≥ 2 are taken as clusters. If no clusters are found at this threshold we fall back to $|\rho| > 0.3$. We select the cluster containing the single most discriminative feature by ANOVA F-score as the candidate set for ablation, with each feature in the cluster ablated individually and independently.

For the cyclic concepts the graph clustering step is omitted and the top 100 features by ANOVA F-score are used directly as the candidate set, with integer cycle labels replacing epoch categories as the grouping variable.

4.6 Superposition Depth Scan

To examine how the temporal representation evolves across the network, we run a layer-wise ablation scan across all 12 layers of GPT2-Small. At each layer l we load the corresponding pretrained SAE, extract residual stream activations at `hook_resid_pre`, and apply the feature clustering procedure described in SECTION REF to identify the candidate feature set. We then establish a baseline temporal correlation $|r|_{\text{base}}$ by encoding the year prompts through the SAE, reconstructing via the decoder, and computing the absolute Pearson correlation between PC1 and numeric year values as described in SECTION REF.

For each feature f_i in the candidate set, we zero its activation across all year prompts simultaneously, setting column i of the ($N_{\text{years}} \times 49,152$) feature activation matrix to zero in SAE space, and recompute the $|r|$ on the modified reconstruction. The drop is:

$$\Delta_i = |r|_{\text{base}} - |r|_{\text{abl},i}$$

A large Δ_i indicates that feature f_i is individually load-bearing for the temporal axis. A near-zero or negative Δ_i indicates negligible contribution or a suppressive feature whose removal improves temporal linearity. PCA is re-fit from scratch on each ablated reconstruction rather than projecting onto a fixed baseline direction, meaning Δ_i measures whether a temporal axis still exists after ablation, not merely whether the ablated reconstruction

aligns with the original one. The test is therefore conservative — a feature only registers as load-bearing if the remaining features cannot reorganise to compensate.

TODO: Talk about the classes created for the plots

Discuss the creation and use of the numerical data without BC/AD

5 Experiments and Results

5.1 Temporal Structure in the Residual Stream

PCA of the raw residual stream at layer 4 reveals that GPT-2 Small organises historical prompts into three distinct clusters in the PC1/PC2 plane and four clusters in three dimensions (Figure 1): BC-era prompts, ancient AD-era prompts, medieval and early modern prompts, and modern prompts, accounting for 73.4% of variance across the top three principal components. PC1 (41.9%) separates ancient from modern prompts while PC2 (20.9%) separates BC from AD within the ancient period.

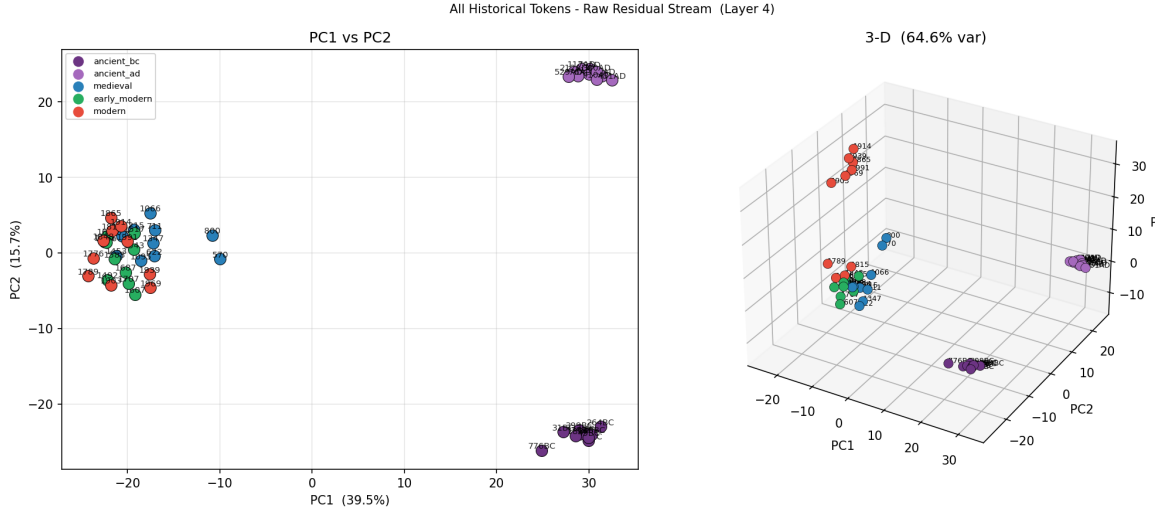


Figure 1: PCA of raw residual stream activations at layer 4. Left: PC1 vs PC2 coloured by epoch category. Right: 3-D scatter

The PC2/PC3 plane reveals a somewhat circular organisation of epoch categories (Figure 2). An algebraic circle fit ($\text{RMSE} = 5.616$) shows the four epoch groups occupying distinct arcs: BC-era prompts cluster on the left arc, ancient AD-era prompts on the right, modern prompts at the top, and medieval through early modern prompts at the bottom. This suggests the PC2/PC3 plane encodes additional temporal structure beyond the PC1 axis, with epoch categories occupying geometrically distinct regions, though the high circle fit error ($\text{RMSE} = 5.616$) indicates the circular structure is approximate rather than precise.

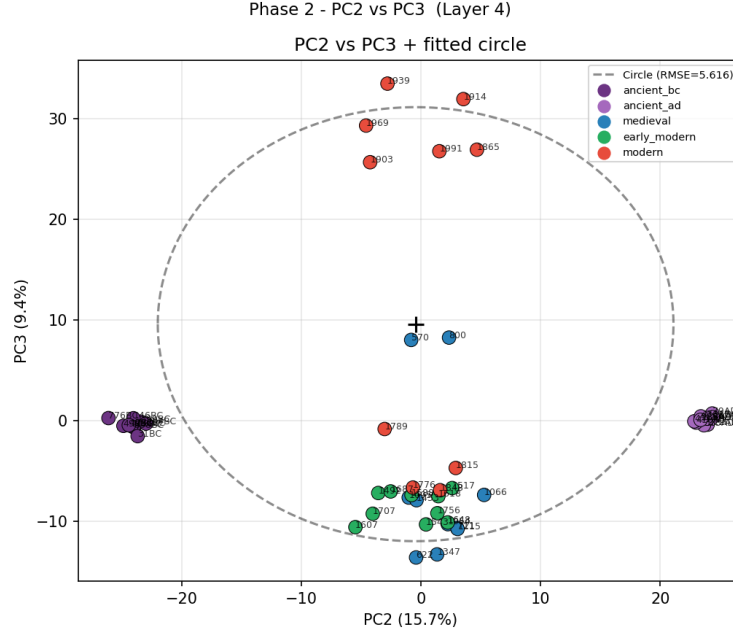


Figure 2: PC2 vs PC3 at layer 4 with algebraic circle fit (RMSE = 5.616).

Angular position on the fitted circle correlates moderately with numeric year ($r = +0.511$, Figure 3), with epoch categories occupying discrete angular bands rather than a smooth progression, confirming the circular structure reflects epoch-level separation rather than a continuous encoding of chronological position.

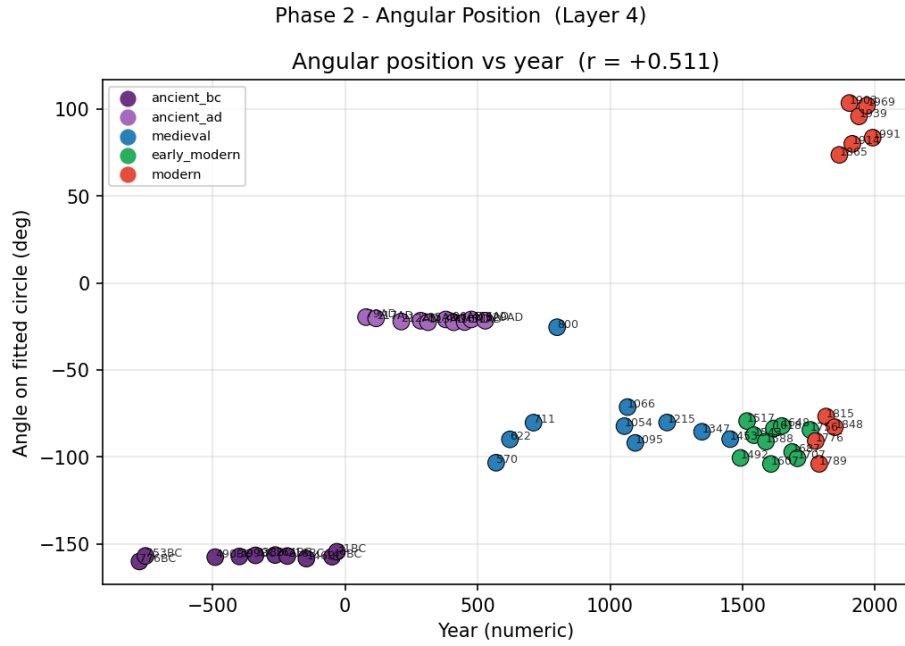


Figure 3: Angular position on fitted circle vs numeric year at layer 4 ($r = +0.511$). Points coloured by epoch category.

5.2 Superposition of Temporal Representations Across Depth

The superposition argument rests on reading $|r|_{\text{base}}$ and $\max_i \Delta_i$ together. A high $|r|_{\text{base}}$ with low $\max_i \Delta_i$ is the fingerprint of distributed superposition — the temporal signal is present and SAE-recoverable, but cannot be attributed to any single feature, as the remaining features compensate collectively for any individual ablation. Conversely, a high $|r|_{\text{base}}$ with high $\max_i \Delta_i$ indicates a concentrated representation where one or a few features are individually load-bearing.

Figure 4 shows the layer-wise depth scan for the original BC/AD corpus. The right panel shows $|r|_{\text{base}}$ remaining stable around 0.85–0.92 across layers 0–9, confirming the temporal signal is consistently present throughout the network. The left panel shows that the $\max_i \Delta_i$ starts high at layers 0–1 (≈ 0.43 , well above the load-bearing threshold of 0.10), then declines monotonically through layers 2–9, falling below the load-bearing threshold by layer 7. This divergence between flat $|r|_{\text{base}}$ and declining $\max_i \Delta_i$ is the superposition signature: the temporal signal is maintained at constant strength but becomes progressively impossible to attribute to any individual SAE feature.

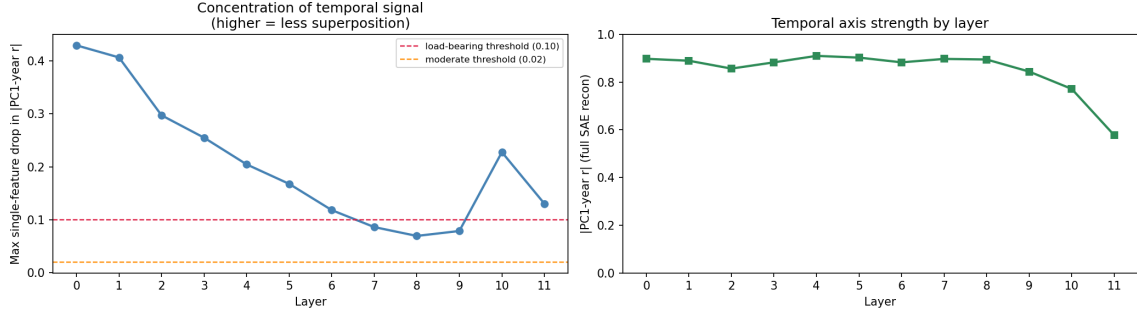


Figure 4: Layer-wise superposition depth scan for the original BC/AD corpus. Left: maximum single-feature ablation drop per layer with load-bearing ($\Delta > 0.10$) and moderate ($\Delta > 0.02$) thresholds. Right: baseline temporal correlation $|r|_{\text{base}}$ per layer.

The full distribution of ablation drops at each layer is shown in Figure 5. At every layer the vast majority of features have negligible drops — the violin bodies are compressed near zero throughout. The upper tails at layers 0–6 represent single outlier load-bearing features, confirming the pattern of one dominant feature with all others negligible rather than a uniform distribution of contributions. From layer 7 onward the tails shrink below the load-bearing threshold, marking the transition to fully distributed superposition.

Two anomalies are worth noting. At layer 10 $\max_i \Delta_i$ spikes back up to ≈ 0.23 after falling near zero at layers 7–9, suggesting a single feature temporarily concentrates the temporal signal during output preparation. At layer 11 $|r|_{\text{base}}$ drops sharply to ≈ 0.58 and the violin shows a wide body of large negative drops — features whose removal improves temporal linearity. Both anomalies are consistent with the model reorganising its residual stream for next-token prediction at the final layers, partially overwriting the stable temporal representation maintained through layers 0–9.

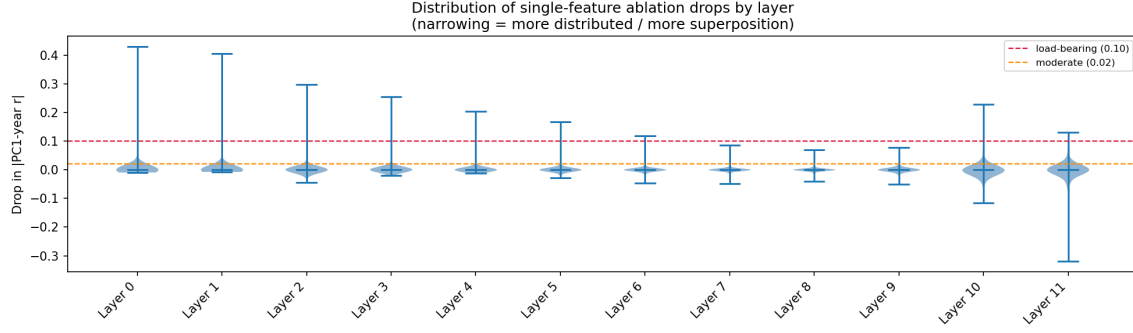


Figure 5: Distribution of single-feature ablation drops by layer for the original BC/AD corpus. Violin bodies compressed near zero at all layers confirm the superposition signature — one outlier load-bearing feature at early layers with all others negligible, transitioning to fully distributed encoding at layers 7–9.

5.3 Surface Token Contamination

Repeating the depth scan on the clean corpus reveals that BC/AD tokens substantially inflate early-layer signals in the original corpus. Figure 6 shows that without BC/AD tokens, PC1 correlation at layer 4 drops from $r = -0.920$ to $r = 0.723$, with PC2 now carrying more temporal information ($r = 0.525$) than before, indicating the temporal signal is more distributed across principal components when surface token identity is unavailable.

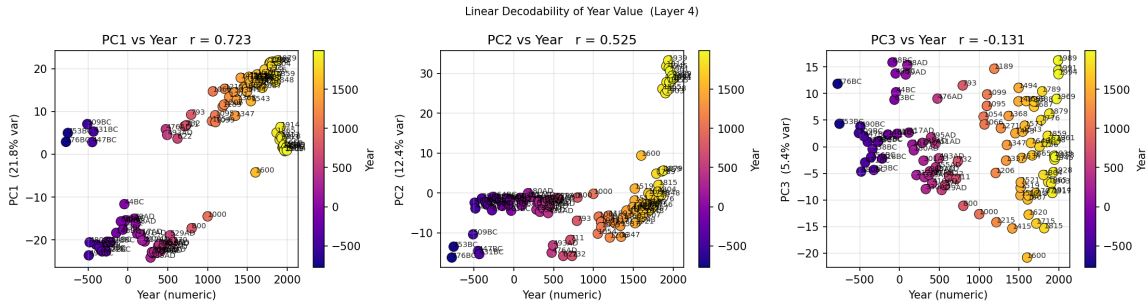


Figure 6: Linear decodability of year value at layer 4 for the clean corpus. PC1 $r = 0.723$, PC2 $r = 0.525$, PC3 $r = -0.131$.

The most striking result is the collapse of $|r|_{\text{base}}$ to near zero at layer 1 (Figure 7, right panel), confirming that the original corpus’s strong early-layer signal was driven by surface token detectors rather than semantic content. Genuine temporal encoding builds monotonically from layer 2, reaching ≈ 0.90 by layers 9–11 and remaining stable through the final layer — unlike the original corpus which degraded at layers 10–11. The concentration profile (left panel) peaks at layers 4–5 rather than layer 0, showing that genuinely load-bearing temporal features emerge at mid-layers. From layer 6 onward both corpora converge to the same fully distributed superposition regime.

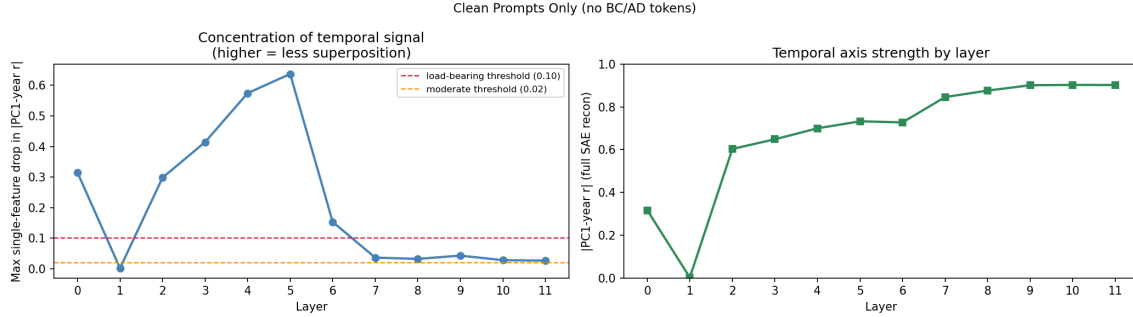


Figure 7: Layer-wise depth scan for the clean corpus. Left: maximum single-feature ablation drop. Right: $|r|_{\text{base}}$ per layer. $|r|_{\text{base}}$ collapses at layer 1 before building monotonically to ≈ 0.90 by layers 9–11.

5.4 Cyclical Concepts: Days of Week and Months of Year

Prior work by Engels et al. [2024] demonstrated that GPT-2 Small encodes days of the week and months of the year as circular representations in the PC2/PC3 plane of the residual stream. We replicate and extend these findings, additionally examining days of the month — a novel cyclic concept with 31 labels that has not previously been studied in this context.

Figure 8 shows the PC2/PC3 geometry for all three concepts at their best layers. Days of the week (layer 7, $r_{\text{angular}} = 0.662$, $\text{RMSE} = 8.201$) and months of the year (layer 7, $r_{\text{angular}} = 0.748$, $\text{RMSE} = 7.070$) both show clear circular organisation, with concepts arranged sequentially around the fitted circle. Days of the month (layer 4, $r_{\text{angular}} = 0.495$, $\text{RMSE} = 2.651$) shows a weaker but present circular structure, with points partially arranged along an arc rather than a full circle. The lower r_{angular} for days of the month is consistent with the increased difficulty of encoding 31 distinct labels compared to 7 or 12.

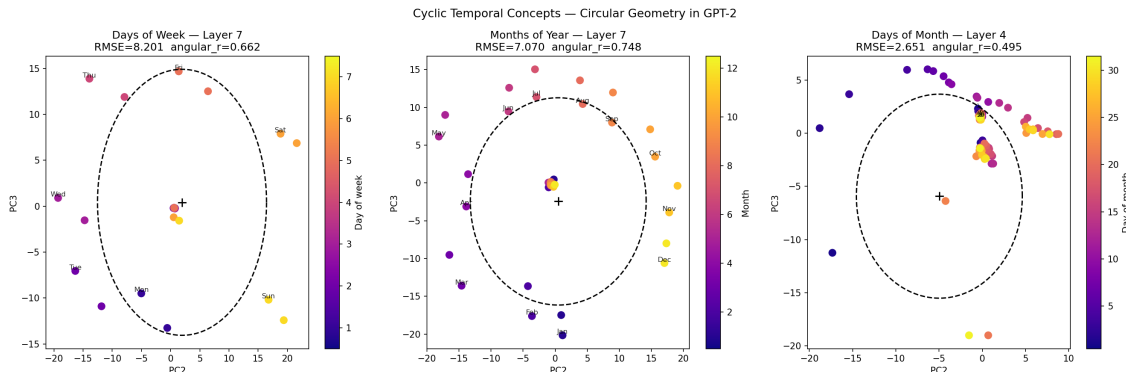


Figure 8: PC2/PC3 geometry for cyclic temporal concepts in GPT-2 Small. Left: days of the week at layer 7 ($r_{\text{angular}} = 0.662$, $\text{RMSE} = 8.201$). Centre: months of the year at layer 7 ($r_{\text{angular}} = 0.748$, $\text{RMSE} = 7.070$). Right: days of the month at layer 4 ($r_{\text{angular}} = 0.495$, $\text{RMSE} = 2.651$). Colour indicates the numeric label for each concept.

5.5 Circular Encoding of Days of the Month

Figure 9 shows the PC2/PC3 geometry for days of the month across five layers and three prompt frames. The all-frames rows reveal that the circular structure is weak and frame-dependent — no layer produces a clean circular arrangement when all three frames are combined.

Examining frames individually tells a clearer story. Frame 1 (simple statement: “*Today is the N th of the month*”) collapses to a tight cluster near the origin at all layers, carrying almost no day-number information in the PC2/PC3 plane. Frame 2 (wraparound sequential: “*The day after the N th is the $(N+1)$ th*”) produces the strongest circular structure, with points arranged along a partial arc that becomes more pronounced at layers 6–8. Frame 3 (positional: “*The N th is the N th day of the month*”) shows linear rather than circular organisation, with day number varying along a single axis rather than wrapping around a circle.

The frame dependence suggests the model does not maintain a single unified circular representation of days of the month. Instead, different prompt framings engage different representational structures.

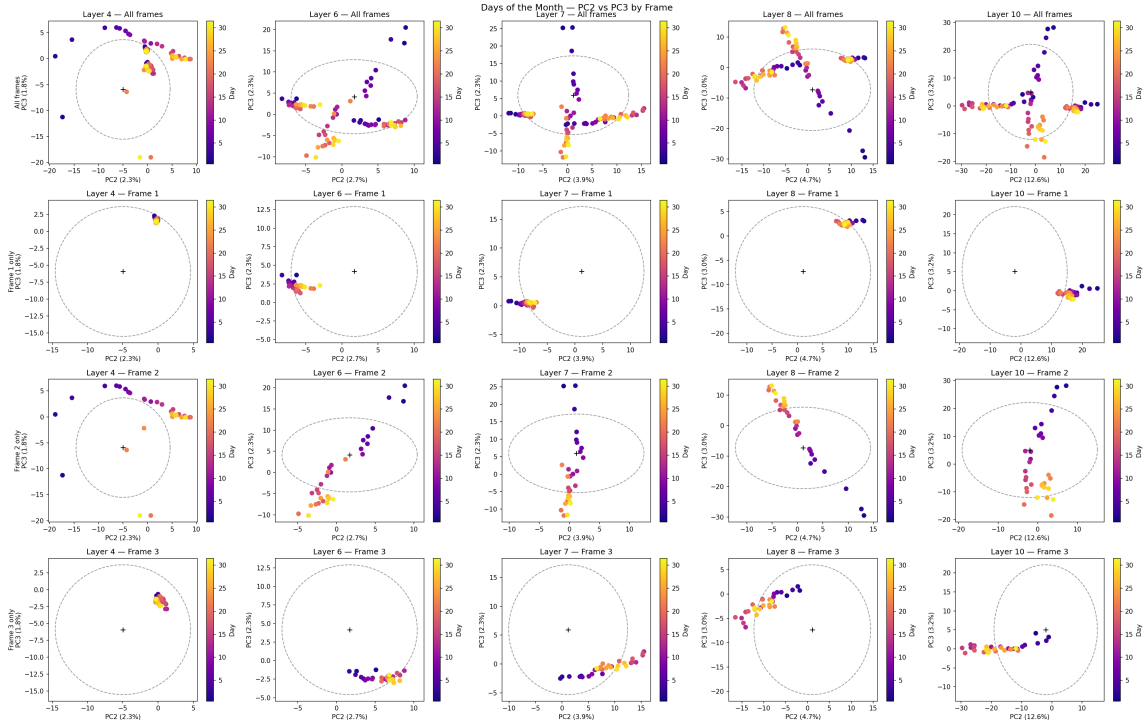


Figure 9: PC2/PC3 geometry for days of the month across five layers (columns) and three prompt frames (rows 2–4), with all frames combined in row 1. Colour indicates day number (1–31).

5.6 Superposition of Circular ideas

TODO (maybe)

6 Discussion

6.1 Linear vs Circular Temporal Encoding

TODO

6.2 The BC/AD Contamination Finding

TODO

6.3 Limitations

Need to think about clustering criteria for which cluster to take for ablating.

Current: Take the cluster that contains the single most discriminative features by ANOVA F-score

Could also try taking the largest cluster.

Need to include cluster sizes for ablation scans

7 Conclusion

TODO

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A Additional Figures